MIL-C-43234C(GL)
30 November 1983
SUPERSEDING
MIL-C-43234B(GL)
5 December 1974

MILITARY SPECIFICATION

# CLOTH, PLAIN WEAVE, ACRYLIC

This specification is approved for use by US Army Natick Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This document covers one type of spun yarn acrylic fiber cloth.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-P-1133 - Packaging of Synthetic Fiber Fabrics

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760 by using the self-addressed Standardization Document improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:58Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Laws and Regulations

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained from without charge from the Federal Trade Commission, Washington, DC 20580.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Copies may be obtained from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Standard sample. The cloth shall match the standard sample for shade, finish and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Acrylic fiber. The yarn shall be acrylic, 3 denier per filament, crimped cut staple.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:58Z
Check the source to verify that this is the current version before use.

3.2.2 Yarn. The yarn shall be carded, drawn and spun into singles or 2 ply yarns for both the warp and the filling.

3.3 Color. The color of the finished cloth shall be Olive Green 106.

3.3.1 Matching. The color of the finished cloth shall match the standard sample under artificial daylight having a color temperature of 7000 ± 500 kelvins and shall be a good approximation to the standard sample under incandescent lamplight at 2850 ± 100 kelvins.

3.3.2 Colorfastness. The finished cloth shall show fastness to laundering, crocking and perspiration equal to or better than the standard sample. When no standard sample is established, the cloth shall show "good" colorfastnes to laundering and perspiration, and shall show an AATCC Chromatic Transference Scale rating for crocking not less than 3.5 when tested as specified in 4.2.3.

3.4 Physical requirements. The finished cloth shall conform to the physical requirements specified in table I when tested as specified in 4.2.3.

TABLE I. Physical requirements

| Weight per square yard (minimum) | Yarns per inch (minimum) | | Breaking strength (minimum) | | Weave |
|---|---|---|---|---|---|
| Ounces | Warp | Filling | Warp | Filling | |
| | | | Pounds | Pounds | |
| 4.8 | 40 | 34 | 75 | 55 | Plain |

3.4.1 Width. The width of the finished cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages when fly-shuttle looms are used and exclusive of selvages when shuttless looms are used.

3.5 Finish. The cloth shall be lightly napped on one side. The degree of character and finish of the cloth shall be equal to that of the standard sample.

3.6 Shrinkage. The cloth shall not shrink more than 4.5 percent in the warp and 3.5 percent in the filling when tested as specified in 4.2.3. The shrinkage process used shall not be identified by name or trademark, either on the cloth, on the ticket or on the package.

3.7 pH. The pH value of the water extract of dyed and finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.2.3.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:58Z
Check the source to verify that this is the current version before use.

3.8 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. The cloth shall be put up in full width rolls as specified in 5.1.

3.9 Fiber identification. Each roll shall be labeled, or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act.

3.10 Workmanship. The finished cloth shall conform to the quality established by this document. The occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 End item examination.

4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the napped side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in roll exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in Section III of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect scales (see 6.4), "F" for slubs and "D" for knots, shall be scored. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points. The sample size



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:58Z
Check the source to verify that this is the current version before use.

shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 35 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 53 points. If one roll exceeds 53 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality. The lot shall be unacceptable if one or more rolls in the second sample exceeds 53 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

* 4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

| | |
|---|---|
| For defects 3 inches or less in any dimension | - one point |
| For defects exceeding 3 inches, but not exceeding 6 inches in any dimension | - two points |
| For defects exceeding 6 inches, but not exceeding 9 inches in any dimension | - three points |
| For defects exceeding 9 inches in any dimension | - four points |

The following defects, when present, shall be scored four points for each yard in which they occur:

Overall uncleanness
Degree of napping not equal to the standard sample
Poor dye penetration, mottled, streaky or cloudy
Width less than specified

4.2.2.2 Length examination.

4.2.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the roll tickets.

4.2.2.3 Shade, finish and appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade, finish and appearance. Any roll in the sample off shade, shaded side to side, side to center, or end to end or any roll that does not have the same finish and appearance as the standard sample shall be cause for rejection of the entire lot represented by the sample.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:58Z
Check the source to verify that this is the current version before use.

4.2.2.4 Roll identification and labeling examination. During the yard-by-yard examination each roll in the sample shall be examined. The lot shall be unacceptable if two or more rolls in the sample contain identification of the shrinkage process by name or trademark on the cloth or ticket, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.2.3 End item testing. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the result of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3 continuous yards, full width of the finished cloth. The sample size (number of sample units) shall be as shown below. The lot shall be unacceptable if one or more units fail to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final result.

| Lot size (yards) | Sample size (sample units) |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and over | 5 |

TABLE III. End item tests

| Characteristic | Requirement paragraph | Test method |
|---|---|---|
| Yarn identification | 3.2.1 | 1/ |
| Colorfastness to: | | |
| Laundering | 3.3.2 | 5610 |
| Crocking | 3.3.2 | 5651 |
| Perspiration | 3.3.2 | 5680 |
| Weight | 3.4 | 5041 |
| Yarns per inch | 3.4 | 5050 |
| Breaking strength | 3.4 | 5100 |
| Weave | 3.4 | Visual 2/ |
| Shrinkage in laundering | 3.6 | 5552 |
| pH | 3.7 | 2811 |

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:58Z
Check the source to verify that this is the current version before use.

2/ One determination shall be made from each sample unit and the result reported as "pass" or "fail".

4.2.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1133.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial. The cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Levels A, B and Commercial. The cloth shall be packaged in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1133.

6. NOTES

6.1 Intended use. The cloth is intended for use as a component in an insulating cap.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Width of cloth required (see 3.4.1).
c. Minimum length, if other than specified (see 3.8).
d. Selection of applicable levels of put-up packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the contracting office issuing the invitation for bids (see 3.1).

* 6.4 Fabric defect scales. Fabric Defect Replica Kits are available from Sears, Roebuck and Company, Department 817 (ATTN: BSC 23-29), Sears Tower, Chicago, IL 60684.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:58Z
Check the source to verify that this is the current version before use.

6.5 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Army - GL

Review activities:

Army - MD
DLA - CT

Preparing activity:

Army - GL

Project No. 8305-A526



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:58Z
Check the source to verify that this is the current version before use.

**INSTRUCTIONS:** In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problems with each item, such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

*(Fold along this line)*

*(Fold along this line)*

DEPARTMENT OF THE ARMY
US Army Natick Research and Development Center
Natick, Massachusetts 01760

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
STRNC-ES

**BUSINESS REPLY MAIL**
FIRST CLASS PERMIT NO. 12062 WASHINGTON D C
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research and Development Center
ATTN: STRNC-ES
Natick, MA 01760

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:58Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-43234C(GL) | Cloth, Plain Weave, Acrylic |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*. _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil – Downloaded: 2026-10-21T20:58Z
Check the source to verify that this is the current version before use.